

LECTURE 3 — DYNAMICAL SYSTEMS (II)

INTRODUCTION TO AUTONOMOUS SYSTEMS · MODULE II

Topics: Observability · Belief state · Flow & well-posedness · Stability · Linearisation · Discretisation

AGENDA

S	TOPIC	KEY IDEAS
3 (cont.)	Partial Observability	Observability matrix · belief state
4	Stability	Flow · equilibria · Lyapunov direct method · linear stability
5	Linearisation	Jacobian linearisation · inverted pendulum
6	Discretisation	Euler instability · ZOH exactness · controller design

Lecture 2 covered §1–§3 up to the Jacobian rank / collinear-beacons example.

S3 (CONT.) · PARTIAL OBSERVABILITY

RECOVERING STATE FROM A SEQUENCE OF OBSERVATIONS



OBSERVABILITY OVER TIME

Even if a single observation loses information, a *sequence* may recover x_0 .

For the linear system $x_{k+1} = Ax_k$, $z_k = Cx_k$, stack n observations:

$$\underbrace{\begin{pmatrix} z_0 \\ \vdots \\ z_{n-1} \end{pmatrix}}_{\text{measurements}} = \underbrace{\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}}_{\mathcal{O} \text{ (observability matrix)}} x_0$$

Definition. (C, A) is **observable** if every initial state x_0 can be uniquely recovered from a finite sequence of outputs $z_0, z_1, \dots$ — i.e. the map $x_0 \mapsto (z_0, \dots, z_{n-1})$ is injective.

OBSERVABILITY: RANK CRITERION AND CONSEQUENCES

Theorem. (C, A) is **observable** $\iff \text{rank}(\mathcal{O}) = n$.

Unobservable subspace: $\ker(\mathcal{O})$ — directions in state space that produce zero output for all time; invisible to any estimator, regardless of algorithm.

Cayley–Hamilton: A^n is a polynomial of $I, A, \dots, A^{n-1}$, so rows beyond CA^{n-1} add no new information — n steps is always sufficient.

Jacobian rank tests *instantaneous* injectivity of h at one step. The observability matrix tests *temporal* injectivity of the joint (A, C) map across n steps.

OBSERVABILITY IN THE NONLINEAR CASE

For $\dot{x} = f(x, a)$, $z = h(x)$, the matrix $\mathcal{O}$ no longer applies. The nonlinear analog uses **Lie derivatives** of h along f (Hermann–Krener Theorem).

What people do in practice: linearise f and h around the current estimate $\hat{x}$ to get $(C, A) = (\partial h / \partial x, \partial f / \partial x)|_{\hat{x}}$, then apply the linear rank criterion. This is a local condition — it can fail far from the operating point — but it is the standard first check and the basis of the Extended Kalman Filter (Module III).

Even when the system is observable, we never recover x_0 exactly — noise makes recovery a distribution, not a point. The right object for a partially observable system is therefore not an estimate but a **belief** $b_t = p(x_t \mid o_{1:t})$.

THE BELIEF – OUR TRUE INPUT

Since o_t alone is insufficient, the policy must act on the **belief**:

$$b_t = p(x_t \mid o_{1:t}, a_{0:t-1})$$

The belief update (Bayes filter):

$$b_t(x_t) \propto \underbrace{p(o_t \mid x_t)}_{\text{measurement}} \cdot \underbrace{\int p(x_t \mid x_{t-1}, a_{t-1}) b_{t-1}(x_{t-1}) dx_{t-1}}_{\text{prediction}}$$

- b_t is a **sufficient statistic** for the full history $o_{1:t}$
- Lectures 4–5: Kalman filter (linear Gaussian exact) and particle filter (nonlinear approximate)

S4 · DYNAMICS AS A FLOW – STABILITY

WHEN DOES A SOLUTION EXIST? WHERE DOES IT GO?



THE FLOW

Definition (Flow). Given $\dot{x} = f(x, a(t))$, the **flow** is the map $\phi : \mathbb{R} \times \mathcal{X} \rightarrow \mathcal{X}$ such that $t \mapsto \phi(t, x_0)$ is the unique solution of the **initial value problem (IVP)** – find $x(t)$ satisfying $\dot{x} = f(x, a)$ with prescribed $x(0) = x_0$:

$$\phi(0, x_0) = x_0, \quad \frac{d}{dt}\phi(t, x_0) = f(\phi(t, x_0), a(t))$$

Two readings:

- **Fix x_0 , vary t** $\rightarrow$ one trajectory (solving the IVP)
- **Fix t , vary x_0** $\rightarrow$ how a whole neighbourhood evolves (stability language)

For an autonomous system $\dot{x} = f(x)$ (no explicit time), the vector field is the same at every moment. So restarting from $\phi(s, x_0)$ at time zero gives the same trajectory as continuing from time s – the dynamics have no memory of the clock. $\phi(t + s, x_0) = \phi(t, \phi(s, x_0))$ – running for s then t equals running for $t + s$.

FLOW OF A LINEAR SYSTEM: DERIVATION

Solve $\dot{x} = Ax, x(0) = x_0$. The **matrix exponential** is defined by the convergent power series:

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{A^2 t^2}{2!} + \dots$$

By analogy with the scalar case $\dot{x} = ax \Rightarrow x(t) = e^{at}x_0$, try:

$$x(t) = e^{At}x_0$$

$$\dot{x} = \frac{d}{dt}e^{At}x_0 = Ae^{At}x_0 = Ax(t) \quad \checkmark$$

$$x(0) = e^{A \cdot 0}x_0 = Ix_0 = x_0 \quad \checkmark$$

By Picard–Lindelöf the solution is unique, so this is the only solution:

$$\boxed{\phi(t, x_0) = e^{At}x_0}$$

WELL-POSEDNESS – PICARD-LINDELÖF

Theorem (Picard–Lindelöf / Cauchy–Lipschitz).

If $f(x, a)$ is **Lipschitz in x** uniformly in a , and $a(t)$ is measurable and bounded, then the IVP has a **unique** solution — the flow $\phi(t, x_0)$ is well-defined.

Lipschitz: $\|f(x, a) - f(x', a)\| \leq L\|x - x'\|$ for all x, x' and **all** $a \in \mathcal{A}$, with a **single** constant L that does not depend on action a .

From C^1 : uniform bound on $\|J_f\|$ gives global Lipschitz. C^1 alone gives only *local* Lipschitz.

Why $a(t)$ measurable (not continuous)? Digital controllers are piecewise constant — measurable but discontinuous. Requiring only measurability covers all realistic control inputs.

WHY LIPSCHITZ MATTERS: NON-UNIQUENESS

If Lipschitz fails, solutions may be non-unique — physics is ambiguous.

Example: $\dot{x} = \sqrt{|x|}$, $x(0) = 0$.

The Lipschitz condition requires $|f(x) - f(0)|/|x| = 1/\sqrt{|x|} \leq L$ near $x = 0$, but this ratio $\rightarrow \infty$ as $x \rightarrow 0$ — no finite L exists.

Consequently two distinct solutions both satisfy the IVP:

- $x(t) \equiv 0$: $\dot{x} = 0 = \sqrt{0}$ ✓
- $x(t) = t^2/4$: $\dot{x} = t/2$ and $\sqrt{x} = \sqrt{t^2/4} = t/2$ ✓

The model cannot predict which trajectory the system follows — the ODE is physically meaningless without Lipschitz.

REMINDER: THE PENDULUM

Setup: mass m , rod l , angle θ from downward vertical, torque u

Lagrangian $L = T - V$:

$$T = \frac{1}{2}ml^2\dot{\theta}^2, \quad V = -mgl \cos \theta$$

Euler–Lagrange $\Rightarrow$ equation of motion:

$$\ddot{\theta} = -\frac{g}{l} \sin \theta + u$$

State–space form $x = (\theta, \dot{\theta})^\top$:

$$\dot{x} = \begin{pmatrix} \dot{\theta} \\ -\frac{g}{l} \sin \theta + u \end{pmatrix}$$

GLOBAL LIPSCHITZ VERIFICATION (PENDULUM)

The Jacobian of f w.r.t. x :

$$J_f(x) = \begin{pmatrix} 0 & 1 \\ -\frac{g}{l} \cos \theta & 0 \end{pmatrix}$$

For an anti-diagonal matrix $\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$, $J^T J = \begin{pmatrix} b^2 & 0 \\ 0 & a^2 \end{pmatrix}$, so the singular values are $|a|$ and $|b|$. Here $a = 1$, $b = -(g/l) \cos \theta$:

$$\|J_f(x)\|_2 = \max(1, (g/l)|\cos \theta|) \leq \max\left(1, \frac{g}{l}\right) =: L \quad \forall x$$

since $|\cos \theta| \leq 1$ everywhere. $\|\cdot\|_2$ is the **spectral norm** (largest singular value).

By the **mean-value theorem** ($\|f(x) - f(y)\| \leq \sup_z \|J_f(z)\|_2 \cdot \|x - y\|$): $\|f(x, a) - f(y, a)\| \leq L\|x - y\|$ for all $x, y \in \mathbb{R}^2$

L is independent of $u(t)$ — the Lipschitz constant is uniform. Picard–Lindelöf applies globally.

LOCAL VS. GLOBAL LIPSCHITZ

Global Lipschitz (single L for all x): unique solution exists for all $t \geq 0$.

Only locally Lipschitz (e.g. C^1 but $\|J_f(x)\| \rightarrow \infty$ as $\|x\| \rightarrow \infty$): a unique solution still exists, but only on some finite time interval $[0, T)$. The solution may **blow up** — $\|x(t)\| \rightarrow \infty$ as $t \rightarrow T$ — and cannot be continued past T .

Global existence in the local case requires an additional argument that keeps trajectories bounded, e.g.:

- An **energy bound**: $V(x) \leq V(x_0)$ prevents escape to infinity
- A **compact invariant set**: the flow stays inside a bounded region by construction

Most physical systems are only locally Lipschitz — global well-posedness is proved via Lyapunov-type arguments, not the Lipschitz constant alone.

EQUILIBRIUM POINTS

Definition. x^* is an **equilibrium** if $f(x^*, a^*) = 0$.

The system at rest: started at x^* , it stays there.

Pendulum ($u = 0$): $f(x^*, 0) = 0$ requires $\dot{\theta}^* = 0$ and $\sin \theta^* = 0$:

EQUILIBRIUM	PHYSICAL POSITION	STABILITY
$(0, 0)$	Hanging down	Stable (centre)
$(\pi, 0)$	Inverted	Unstable (saddle)

Controlled equilibria: any angle θ^* with constant torque $u^* = \frac{g}{l} \sin \theta^*$

Remark (physical pendulum): for rigid body with I_{pivot} , equilibria are the same two points — $\sin \theta^* = 0$ is unaffected by replacing ml^2 with I_{pivot} .

WHY GRAPHICAL ANALYSIS?

Most nonlinear ODEs **cannot be solved in closed form**.

Even the pendulum $\ddot{\theta} = -\frac{g}{l} \sin \theta$ has no elementary solution — the exact period requires an **elliptic integral**:

$$T = 4\sqrt{\frac{l}{g}} \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - k^2 \sin^2 \psi}}, \quad k = \sin \frac{\theta_0}{2}$$

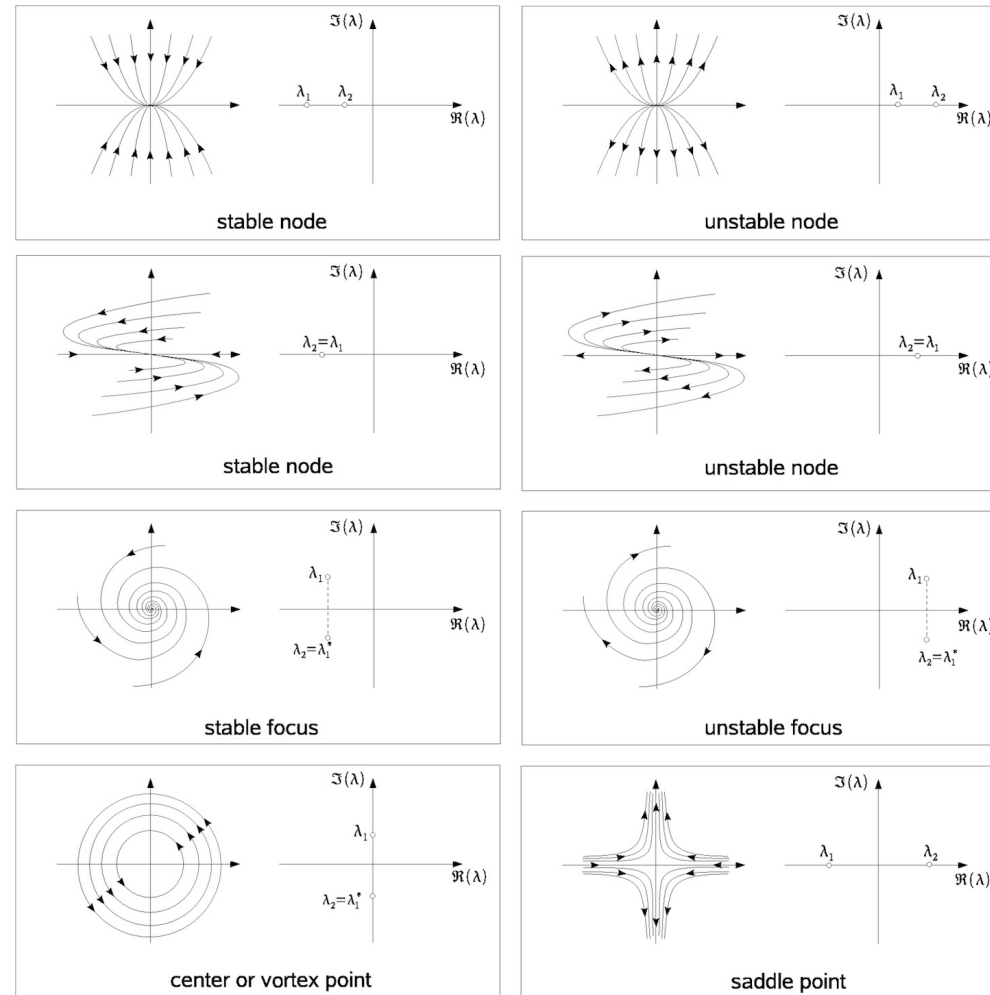
Analytic intractability is the norm for real systems, not the exception. So instead of solving, we **extract qualitative information graphically**.

A **phase portrait** answers questions that matter in engineering:

- Will my robot fall? i.e. $\lim_{t \rightarrow \infty} \theta(t) = ?$
- Which initial conditions lead to safe operation vs. failure?
- Where are the equilibria, and are they stable?

Moral: When you cannot compute the trajectory, compute the geometry of the flow.

CLASSIFYING FIXED POINTS



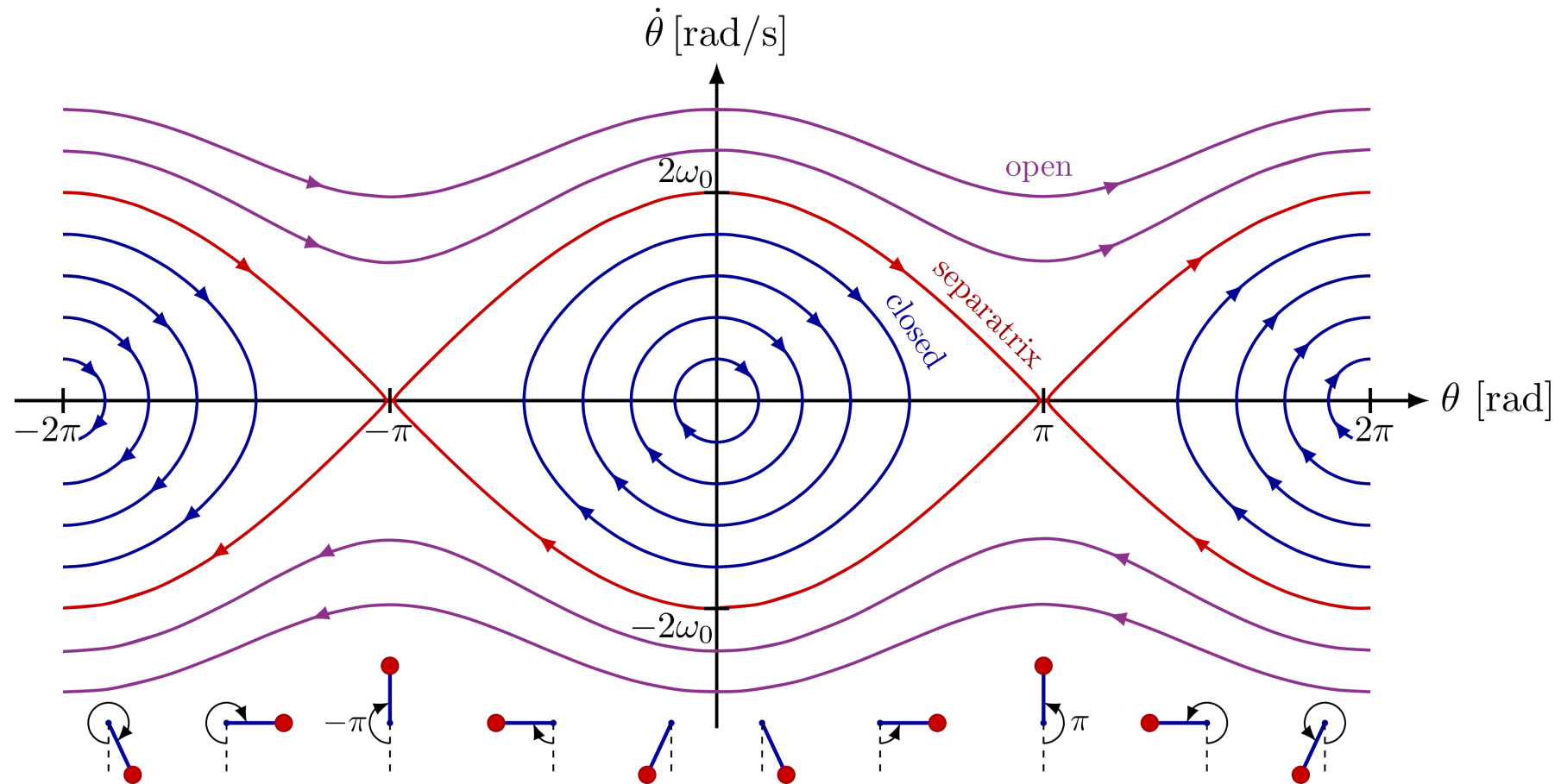
CLASSIFYING FIXED POINTS: PENDULUM

Linearise at each equilibrium x^* : eigenvalues of $A = J_f(x^*)$ determine the local geometry.

Pendulum ($u = 0$): $J_f = \begin{pmatrix} 0 & 1 \\ -\frac{g}{l} \cos \theta & 0 \end{pmatrix}$, characteristic eq. $\lambda^2 + \frac{g}{l} \cos \theta^* = 0$

EQUILIBRIUM	$\cos \theta^*$	A	TYPE
$(0, 0)$ – hanging	$+1$	$\pm i\sqrt{g/l}$	Centre
$(\pi, 0)$ – inverted	-1	$\pm \sqrt{g/l}$	Saddle

PHASE PORTRAIT – PENDULUM



https://tikz.net/dynamics_phaseportrait/

PHASE PORTRAIT – PENDULUM

Energy $E = \frac{1}{2}ml^2\dot{\theta}^2 - mgl \cos \theta$ is conserved ($\dot{E} = 0$). Trajectories = level sets of E .

REGIME	BEHAVIOUR
$E < mgl$	Libration — closed orbits around $(0, 0)$
$E = mgl$	Separatrix — arrives at saddle $(\pm\pi, 0)$ in ∞ time
$E > mgl$	Rotation — full revolutions

The separatrix is the boundary between qualitatively different behaviours — crossing it is the central challenge in swing-up control.

WHAT HAPPENS TO THE PHASE PORTRAIT IF WE ADD DAMPING?



LYAPUNOV STABILITY – DEFINITIONS

Definition (Lyapunov stable). x^* is Lyapunov stable if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \|x_0 - x^*\| < \delta \Rightarrow \|x(t) - x^*\| < \varepsilon \quad \forall t \geq 0$$

Trajectories starting close stay close. Says nothing about convergence.

Definition (Asymptotically stable, AS). Lyapunov stable and $\exists \delta > 0$ s.t. $\|x_0 - x^*\| < \delta \Rightarrow x(t) \rightarrow x^*$.

LYAPUNOV STABILITY HIERARCHY

NAME	MEANING
Lyapunov stable	Close trajectories stay close
Asymptotically stable (AS)	Lyapunov stable + $x(t) \rightarrow x^*$
Marginally stable	Lyapunov stable but NOT AS
Globally AS (GAS)	AS for all $x_0 \in \mathcal{X}$
Exponentially stable	$\ x(t) - x^*\ \leq c e^{-\lambda t} \ x_0 - x^*\ $, some $c, \lambda > 0$
Unstable	Not Lyapunov stable

Pendulum at $(0, 0)$ without damping: marginally stable — orbits, energy conserved, never converges.

Add $b\dot{\theta}$ damping: $\longrightarrow$ asymptotically stable.

LYAPUNOV'S DIRECT METHOD

Theorem. If a C^1 function $V : \mathcal{X} \rightarrow \mathbb{R}$ satisfies:

1. $V(x^*) = 0$
2. $V(x) > 0$ for $x \neq x^*$ near x^*
3. $\dot{V}(x) = \nabla V \cdot f(x) \leq 0$ near x^*

then x^* is **Lyapunov stable**.

Strict $\nabla V \cdot f(x) < 0$ for $x \neq x^*$ $\Rightarrow$ **asymptotically stable**.

Radially unbounded ($V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$) + conditions hold on all of $\mathcal{X}$ (not just near x^*) $\Rightarrow$ **GAS** (globally asymptotically stable).

Key insight: No need to solve the ODE — only need a "generalised energy" V .

Why does it work? Level sets $\{V \leq c\}$ are positively invariant when $\dot{V} \leq 0$ — once inside, the trajectory cannot escape.

EXAMPLE: DAMPED PENDULUM (LYAPUNOV)

System: $\dot{\theta}_1 = \theta_2, \dot{\theta}_2 = -\sin \theta_1 - b\theta_2$ ($b > 0$)

Candidate — physical energy shifted so $V(x^*) = 0$. With $m = l = g = 1$ and mass height $y = -\cos \theta_1$:

$$E = \underbrace{-\cos \theta_1}_{\text{potential}} + \underbrace{\frac{1}{2}\theta_2^2}_{\text{kinetic}}, \quad E(0, 0) = -1$$

Subtract $E(x^*) = -1$ to enforce $V(x^*) = 0$:

$$V = E - E(0, 0) = \underbrace{(1 - \cos \theta_1)}_{\geq 0} + \underbrace{\frac{1}{2}\theta_2^2}_{\geq 0} \quad V(0, 0) = 0 \checkmark$$

$$\dot{V} = \sin \theta_1 \cdot \theta_2 + \theta_2(-\sin \theta_1 - b\theta_2) = -b\theta_2^2 \leq 0 \checkmark$$

$\dot{V} = 0$ only on $\{\theta_2 = 0\}$, but the system cannot stay there (unless at equilibrium).

Therefore, origin is **asymptotically stable**.

FINDING A LYAPUNOV FUNCTION – 4 STRATEGIES

STRATEGY	SYSTEM CLASS	HOW
1. Physical energy	Mechanical, electrical	$V = E - E(x^*)$
2. Lyapunov equation	Linear $\dot{x} = Ax$	Pick any $Q \succ 0$; solve $A^\top P + PA = -Q$ for P
3. From linearisation	Nonlinear (local)	Use P from linearisation at x^* ; certifies AS locally
4. Sum-of-squares (SOS)	Polynomial f	Search for polynomial V via semidefinite programming (SDP) — a convex optimisation over positive-semidefinite matrices (SOSTOOLS, Drake)

Converse theorem (Massera): If x^* is AS, a smooth V *always exists* — but the theorem does not construct it.

LINEAR STABILITY – EIGENVALUE CRITERION

Solution of $\dot{x} = Ax$: $x(t) = e^{At}x_0$

Each mode $e^{\lambda_i t}$ decays $\iff \operatorname{Re}(\lambda_i) < 0$

Theorem (Continuous Linear Stability).

- **AS** $\iff$ all $\operatorname{Re}(\lambda_i) < 0$
- **Marginally stable** $\iff$ all $\operatorname{Re}(\lambda_i) \leq 0$ and boundary eigenvalues **semisimple**
- **Unstable** otherwise

Semisimple at boundary: algebraic multiplicity = geometric multiplicity (no Jordan blocks).

Without it: polynomial growth $\|x(t)\| \sim t^{m-1} \rightarrow \infty$ even with $\operatorname{Re}(\lambda) = 0$.

QUADRATIC LYAPUNOV FOR LINEAR SYSTEMS

If all $\operatorname{Re}(\lambda_i) < 0$, pick any $Q \succ 0$ and solve the **algebraic Lyapunov equation**:

$$PA + A^\top P = -Q$$

A unique $P \succ 0$ exists $\iff A$ is **Hurwitz** (all eigenvalues have $\operatorname{Re}(\lambda_i) < 0$). Then:

$$V(x) = x^\top P x, \quad \dot{V} = x^\top (PA + A^\top P)x = -x^\top Q x < 0$$

This certifies **GAS** — and gives an explicit Lyapunov function for the linear system.

S5 · LINEARISATION

TURNING NONLINEAR INTO LINEAR – LOCALLY



JACOBIAN LINEARISATION

At equilibrium (x^*, a^*) , Taylor-expand f :

$$\dot{\delta x} = \underbrace{\left. \frac{\partial f}{\partial x} \right|_{(x^*, a^*)}}_A \delta x + \underbrace{\left. \frac{\partial f}{\partial a} \right|_{(x^*, a^*)}}_B \delta a + O(\|\delta x\|^2)$$

Drop higher-order terms: $\dot{\delta x} = A \delta x + B \delta a$

- $A \in \mathbb{R}^{n \times n}$ — open-loop dynamics matrix
- $B \in \mathbb{R}^{n \times m}$ — input matrix
- **Valid only in a neighbourhood of x^***

EXAMPLE: INVERTED PENDULUM

Equilibrium: $x^* = (\pi, 0), u^* = 0$

$$f(x, u) = \begin{pmatrix} x_2 \\ -\frac{g}{l} \sin x_1 + u \end{pmatrix} \quad A = \left. \frac{\partial f}{\partial x} \right|_{(\pi, 0)} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{l} \cos \pi & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{g}{l} & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Eigenvalues of A : $\lambda = \pm \sqrt{g/l}$

One positive real eigenvalue $\Rightarrow$ **unstable** (confirmed without Lyapunov)

S6 · DISCRETISATION

FROM CONTINUOUS ODES TO DIGITAL CONTROLLERS



WHY DISCRETISE?

Controllers and estimators run on **digital hardware** at rate $1/\Delta t$.

Goal: replace $\dot{x} = f(x, a)$ with a map $x_{k+1} = g(x_k, a_k)$

Two very different uses:

USE	WHAT WE NEED	METHOD
Simulation	Accurate trajectory	RK4, adaptive RK45
Controller design	Exact plant model	Zero-Order Hold (ZOH)

EULER METHOD – DERIVATION

$$x_{k+1} = x_k + \Delta t f(x_k, a_k)$$

Where it comes from. Taylor-expand the exact solution; use $\dot{x} = f(x)$ and $\ddot{x} = \frac{d}{dt}f(x) = J_f(x)\dot{x} = J_f(x)f(x)$:

$$x(t_k + \Delta t) = x_k + \Delta t \underbrace{\dot{x}(t_k)}_{f(x_k)} + \frac{\Delta t^2}{2} \underbrace{\ddot{x}(t_k)}_{J_f(x_k)f(x_k)} + O(\Delta t^3)$$

Euler keeps only the Δt term. The dropped piece is the **local truncation error** (LTE):

$$\text{LTE} = \frac{\Delta t^2}{2} J_f(x_k) f(x_k) + O(\Delta t^3) = O(\Delta t^2)$$

Over $T/\Delta t$ steps: **global error** $\approx \frac{T}{\Delta t} \cdot O(\Delta t^2) = O(\Delta t)$ – **first-order method**.

EULER METHOD – LINEAR SYSTEMS

For $\dot{x} = Ax$, substituting into the Euler formula:

$$x_{k+1} = x_k + \Delta t Ax_k = (I + \Delta t A) x_k$$

$$A_d^{\text{Euler}} = I + \Delta t A$$

If λ is an eigenvalue of A , the corresponding discrete eigenvalue is $1 + \Delta t \lambda$.

Euler is taught because it makes stability analysis transparent.

It is **not** the method of choice for actual simulation.

DISCRETE LINEAR STABILITY

Theorem (Discrete Linear Stability).

$x_{k+1} = A_d x_k$ is **asymptotically stable** $\iff$ all eigenvalues μ_i satisfy $|\mu_i| < 1$ – inside the **unit circle** in $\mathbb{C}$.

Solution: $x_k = A_d^k x_0$. Each mode $\mu_i^k \rightarrow 0$ iff $|\mu_i| < 1$.

STABILITY	CONTINUOUS	DISCRETE
Asymp. stable	all $\operatorname{Re}(\lambda_i) < 0$	all $ \mu_i < 1$
Marginally stable	all $\operatorname{Re}(\lambda_i) \leq 0$, boundary semisimple	all $ \mu_i \leq 1$, boundary semisimple
Unstable	some $\operatorname{Re}(\lambda_i) > 0$, or non-semisimple	some $ \mu_i > 1$, or non-semisimple

Semisimple at boundary: algebraic multiplicity = geometric multiplicity (no Jordan blocks). Without it, polynomial growth $t^{m-1} \rightarrow \infty$ even though $\operatorname{Re}(\lambda) = 0$ or $|\mu| = 1$.

EULER BREAKS A STABLE SYSTEM

Pendulum at $(0, 0)$: $\lambda = \pm i\omega, \omega = \sqrt{g/l}$ – true system is **marginally stable** (closed orbits, energy conserved)

$$\text{Euler gives } A_d^{\text{Euler}} = I + J_f \Delta t = \begin{pmatrix} 1 & \Delta t \\ -\frac{g}{l} \Delta t & 1 \end{pmatrix}$$

$$\det(A_d^{\text{Euler}}) = 1 + \frac{g}{l} \Delta t^2 > 1 \quad \forall \Delta t > 0$$

Real matrix $\Rightarrow$ conjugate eigenvalue pair $\mu, \bar{\mu} \Rightarrow |\mu| = |\bar{\mu}|$

$\Rightarrow |\mu|^2 = \det > 1 \Rightarrow$ **both eigenvalues outside the unit circle – unstable**

Euler predicts exponential growth for any step size – even though the pendulum oscillates stably.

WHAT TO USE FOR SIMULATION

SITUATION	METHOD	WHY
Smooth nonlinear	RK4	$O(\Delta t^4)$, large stability region
Unknown stiffness	Adaptive RK45 (<code>ode45</code>)	Auto-adjusts Δt
Stiff systems	Implicit (backward Euler, Crank–Nicolson)	Unconditionally stable

Explicit vs. implicit. Explicit methods (Euler, RK4) compute x_{k+1} directly from x_k — cheap per step but require small Δt for stability. Implicit methods solve $x_{k+1} = x_k + \Delta t f(x_{k+1})$ at each step — more expensive but unconditionally stable.

Stiffness — eigenvalues with very different magnitudes, e.g. $\lambda_1 = -1000$ and $\lambda_2 = -1$. Explicit methods must step at the timescale of the fastest mode ($\sim 1/1000$) to stay stable, even if you only care about the slow one. Implicit methods have no such restriction.

Why not use simulation methods for controller design?

Approximate model $\Rightarrow$ controller is optimal for the *wrong plant* $\Rightarrow$ stability guarantee may not transfer.

THE ZOH PIPELINE

Zero-Order Hold (ZOH) — exact discretization of a linear system under the assumption that the digital controller holds its command constant between samples: $u(t) = u_k$ on $[t_k, t_k + \Delta t)$.

$$\underbrace{\dot{x} = f(x, u)}_{\text{nonlinear}} \xrightarrow{\text{linearise (§5)}} \underbrace{\dot{\delta x} = A \delta x + B \delta u}_{\text{LTI}} \xrightarrow{\text{ZOH}} \underbrace{\delta x_{k+1} = A_d \delta x_k + B_d \delta u_k}_{\text{discrete LTI}}$$

The discrete LTI model feeds:

- **LQR** — optimal discrete-time state feedback (Lecture 7)
- **Kalman filter** — optimal discrete-time estimator (Lecture 5)
- **MPC** — rolling short-horizon optimiser (Lecture 8)

ZOH is the bridge between continuous physics and digital control.

ZOH DERIVATION (1/2)

Hold $u(t) = u_k$ constant on $[t_k, t_k + \Delta t)$. Solve $\dot{x} = Ax + Bu_k$ exactly.

Integrating factor. Define $g(s) = e^{-A(s-t_k)}x(s)$. Differentiate:

$$\frac{dg}{ds} = -Ae^{-A(s-t_k)}x + e^{-A(s-t_k)}\frac{dx}{ds} = e^{-A(s-t_k)}\underbrace{(-Ax + Ax + Bu_k)}_{Bu_k} = e^{-A(s-t_k)}Bu_k$$

Integrate from t_k to $t_k + \Delta t$:

$$g(t_k + \Delta t) - g(t_k) = \int_{t_k}^{t_k + \Delta t} e^{-A(s-t_k)}Bu_k ds$$

ZOH DERIVATION (2/2)

Substitute back: $g(t_k) = x_k$, $g(t_k + \Delta t) = e^{-A\Delta t}x_{k+1}$. Multiply by $e^{A\Delta t}$:

$$x_{k+1} = e^{A\Delta t}x_k + \int_{t_k}^{t_k+\Delta t} e^{A(t_k+\Delta t-s)} Bu_k ds$$

Change of variable $\tau = t_k + \Delta t - s$ ($d\tau = -ds$, limits flip):

$$x_{k+1} = \underbrace{e^{A\Delta t}}_{A_d} x_k + \underbrace{\left(\int_0^{\Delta t} e^{A\tau} d\tau \right)}_{B_d} Bu_k$$

This is the **exact** ODE solution — not an approximation.

COMPUTING (A_D, B_D) – INVERTIBLE CASE

We need $A_d = e^{A\Delta t}$ and $B_d = \left(\int_0^{\Delta t} e^{A\tau} d\tau \right) B$.

When A is invertible. The antiderivative of $e^{A\tau}$ is $A^{-1}e^{A\tau}$ (since $\frac{d}{d\tau} A^{-1}e^{A\tau} = e^{A\tau}$):

$$\int_0^{\Delta t} e^{A\tau} d\tau = A^{-1}e^{A\tau} \Big|_0^{\Delta t} = A^{-1}(e^{A\Delta t} - I) \implies B_d = A^{-1}(e^{A\Delta t} - I)B$$

COMPUTING (A_D, B_D) – GENERAL CASE

Block exponential. ZOH holds u constant, so treat it as a state with $\frac{du}{d\tau} = 0$. The augmented system:

$$\frac{d}{d\tau} \begin{pmatrix} x \\ u \end{pmatrix} = \underbrace{\begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix}}_M \begin{pmatrix} x \\ u \end{pmatrix} \implies \begin{pmatrix} x(\Delta t) \\ u(\Delta t) \end{pmatrix} = e^{M\Delta t} \begin{pmatrix} x_0 \\ u_0 \end{pmatrix}$$

Reading the top block: $x(\Delta t) = [e^{M\Delta t}]_{11} x_0 + [e^{M\Delta t}]_{12} u_0$. But we know $x(\Delta t) = A_d x_0 + B_d u_0$, so:

$$e^{M\Delta t} = \begin{pmatrix} A_d & B_d \\ 0 & I \end{pmatrix}$$

Payoff: just call `expm(M * Δt)` – top-left is A_d , top-right is B_d . No A^{-1} needed, works for any A .

ZOH STABILITY PRESERVATION

If λ_i are eigenvalues of A , then eigenvalues of $A_d = e^{A\Delta t}$ are $e^{\lambda_i \Delta t}$:

$$|e^{\lambda_i \Delta t}| = e^{\operatorname{Re}(\lambda_i) \Delta t} \begin{cases} < 1 & \text{if } \operatorname{Re}(\lambda_i) < 0 \\ = 1 & \text{if } \operatorname{Re}(\lambda_i) = 0 \\ > 1 & \text{if } \operatorname{Re}(\lambda_i) > 0 \end{cases}$$

Key result: ZOH preserves stability for **all** $\Delta t > 0$.

A continuously stable LTI system gives a discretely stable ZOH system — no step-size restriction.

Pendulum at $(0, 0)$: $\lambda = \pm i\omega \Rightarrow |e^{\pm i\omega \Delta t}| = 1$ — exactly on the unit circle, marginal stability preserved ✓

STABILITY ANALYSIS IN CONTROLLER DESIGN

After ZOH we have the discrete plant (A_d, B_d) . Apply **linear state feedback** $u_k = Kx_k$, where $K \in \mathbb{R}^{m \times n}$ is a gain matrix to be designed. Substituting into $x_{k+1} = A_dx_k + B_d u_k$:

$$x_{k+1} = \underbrace{(A_d + B_d K)}_{\text{closed-loop } A_{cl}} x_k \quad \text{is stable}$$

Requires $\text{eig}(A_d + B_d K)$ **inside the unit circle**.

PLANT	A_D EIGENVALUES	CONTROLLER TASK
Stable	Inside disk	Verify closed-loop still stable
Unstable (e.g. inverted pendulum)	Outside disk	Must move them inside

Why ZOH exactness matters:

(A_d, B_d) exact $\Rightarrow$ discrete stability/optimality guarantee transfers to real sampled system.

With Euler/RK4: approximate model $\Rightarrow$ guarantee may not transfer.

WHAT'S NEXT

Lecture 4 — Probability and Optimisation

- Bayesian inference on the belief b_t
- Gradient-based policy optimisation
- How the Markov structure from today enables tractable inference